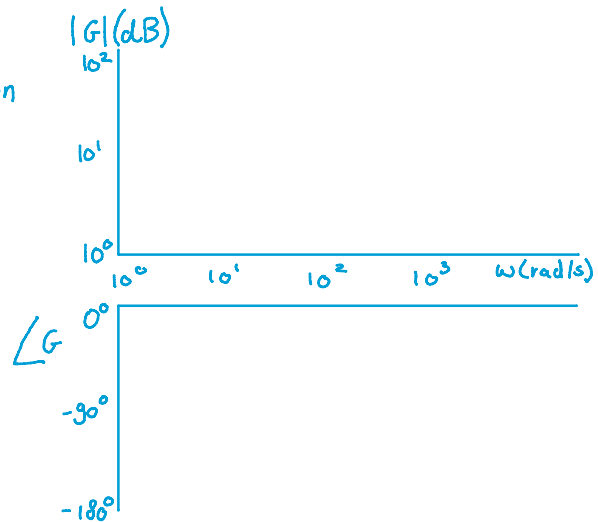


3 - Frequency response - Type 1 and type 2 building blocks

Frequency response analysis

- Frequency response analysis
 - ↳ System identification
 - ↳ Controller design
 - ↳ visualisation with **Bode Plots**



- Steady state behaviour
 - ↳ only sinusoidal inputs
 - ↳ Math operations
 - addition/subtraction
 - multiplication/division
 - Integrating / derivating
- } change in
magnitude &
phase, not frequency

- Polar form $G(j\omega_0) = \underbrace{M}_{\text{phase}} e^{j\phi}$ } magnitude

• Composite system

$$G(j\omega) = \frac{M_1 e^{j\phi_1} M_2 e^{j\phi_2}}{M_3 e^{j\phi_3} M_4 e^{j\phi_4} M_5 e^{j\phi_5}} = \frac{M_1 M_2}{M_3 M_4 M_5} e^{j(\phi_1 + \phi_2 - \phi_3 - \phi_4 - \phi_5)}$$

$$|G(j\omega)| = \log(M_1 + M_2 - M_3 - M_4 - M_5)$$

- $s = j\omega$ Frequency dependent $K G(j\omega) = K_0(j\omega)^n \frac{(j\omega\tau_1 + 1)(j\omega\tau_2 + 1) \dots}{(j\omega\tau_a + 1)(j\omega\tau_b + 1) \dots}$
 - Three classes:
 1. $K_0(j\omega)^n$
 2. $(j\omega\tau + 1)^{\pm 1}$
 3. $\left[\left(\frac{j\omega}{\omega_n} \right)^2 + 2\zeta \frac{j\omega}{\omega_n} + 1 \right]^{\pm 1}$
- Break point
Frequencies

Class 1: singularities at the origin

$$K G(s) = K \frac{(s-z_1)(s-z_2)\dots}{(s-p_1)(s-p_2)\dots} \xleftrightarrow{s=j\omega} \underbrace{K_0(j\omega)^n}_{\text{Class 1}} \frac{(j\omega\tau_1+1)(j\omega\tau_2+1)\dots}{(j\omega\tau_a+1)(j\omega\tau_b+1)\dots}$$

$K \neq K_0$!

degree of poles/zeros

At the origin:

- Pole $1/s$ or s^{-1} } integration } Laplace
- Zero s or s^1 } derivation }

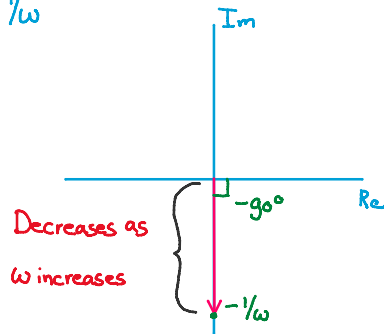
Illustration

$$\frac{1}{j\omega} = 0 + \frac{1}{\omega} \underbrace{\frac{1}{j} \frac{j}{j}} = 0 - \frac{1}{\omega} j$$

$$\frac{1}{j-1} \cdot \frac{j-1}{j-1} = \frac{j-1}{-1} = -j$$

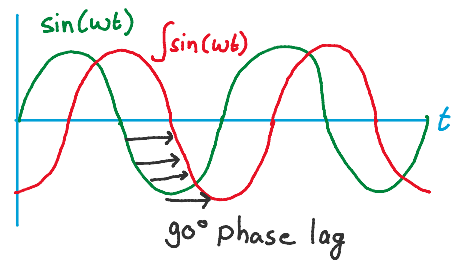
Real: 0

Imag: $-1/\omega$



$$\sin(\omega t) \xrightarrow{1/s} \int \sin(\omega t) dt = -\frac{1}{\omega} \cos(\omega t) = -\frac{1}{\omega} \sin(\omega t + \phi)$$

$\phi = -90^\circ$

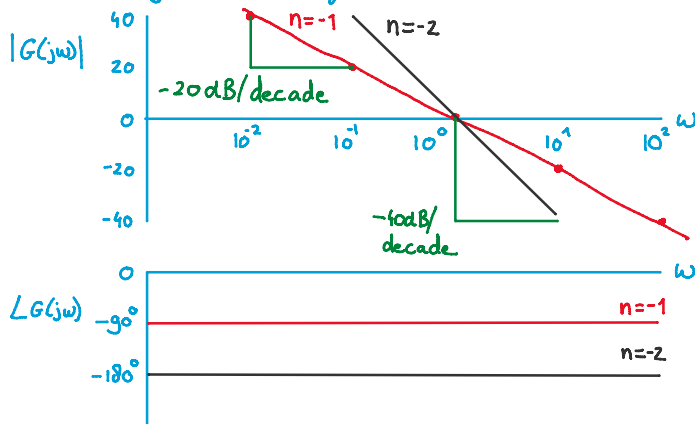


Class 1: in bode form $1/s$

$\omega = 1$: gain = 1 $20 \log(1) = 0$

$\omega = 10$ gain = $1/10$ $20 \log(1/10) = -20$

$\omega = 100$ gain = $1/100$ $20 \log(1/100) = -40$

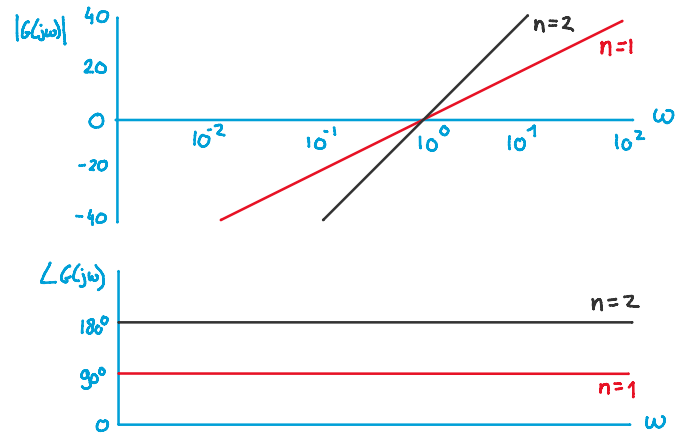


Bode gain-phase relationship

Gain: $n \times 20 \text{ dB/decade}$

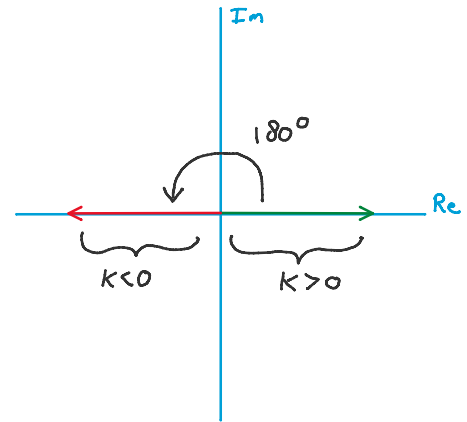
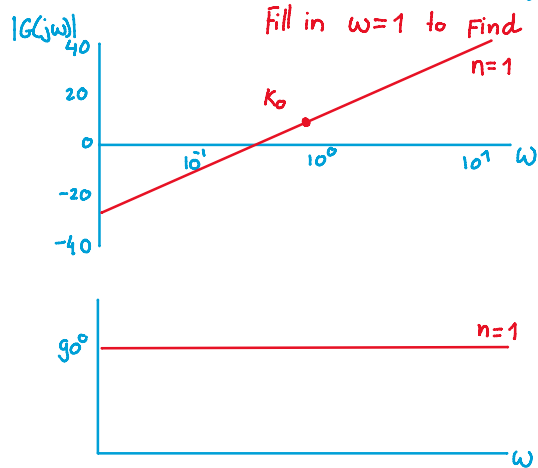
Phase: $n \times 90^\circ$

S^n



$K_o(j\omega)^n$ $\log K_o |G(j\omega)^n| = \log K_o + n \log(j\omega)$

Fill in $\omega=1$ to Find K_o



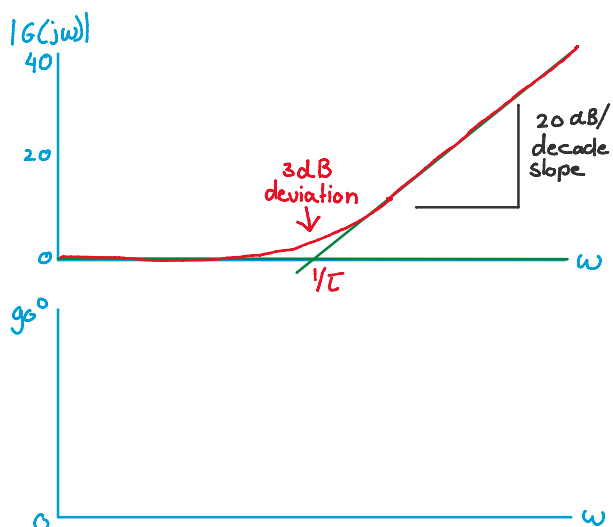
Class 2: First order terms

$\begin{cases} +1 & \text{Zero} \\ -1 & \text{Pole} \end{cases}$

$$(j\omega\tau + 1)^{\pm 1}$$

$$\begin{cases} \omega\tau \ll 1, & j\omega\tau + 1 \cong 1 \rightarrow 0 \text{ dB} \\ \omega\tau \gg 1, & j\omega\tau + 1 \cong j\omega\tau \rightarrow \pm 20 \text{ dB/decade} \end{cases}$$

Break point frequency τ
 $\omega = 1/\tau$ Break point



Pole

$$\frac{1}{j\omega\tau + 1} \cdot \frac{1-j\omega\tau}{1-j\omega\tau} = \frac{1-j\omega\tau}{1+\omega^2\tau^2}$$

$$\begin{aligned} \text{Re: } & 1/(1+\omega^2\tau^2) \\ \text{Im: } & -\omega\tau/(1+\omega^2\tau^2) \end{aligned}$$

Zero

$$1+j\omega\tau$$

$$\begin{aligned} \text{Re: } & 1 \\ \text{Im: } & \omega\tau \end{aligned}$$

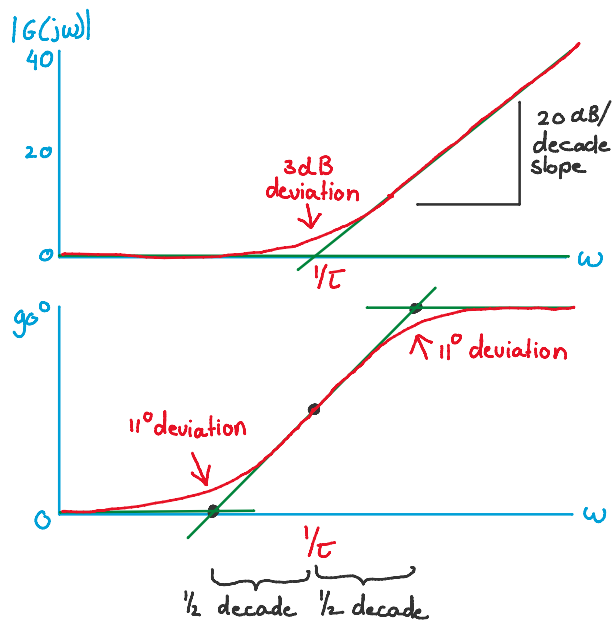
$$\begin{aligned} \text{Gain: } & \left[\left(\frac{1}{1+\omega^2\tau^2} \right)^2 + \left(\frac{-\omega\tau}{1+\omega^2\tau^2} \right)^2 \right]^{1/2} = \left(\frac{1+\omega^2\tau^2}{(1+\omega^2\tau^2)^2} \right)^{1/2} \\ & = -20 \log(\sqrt{1+\omega^2\tau^2}) \text{ in decibels} \end{aligned}$$

$$\text{Phase: } \arctan(\text{Im}/\text{Re}) = \arctan(-\omega\tau)$$

$$\text{Gain: } \sqrt{1+\omega^2\tau^2} = 20 \log(\sqrt{1+\omega^2\tau^2})$$

$$\text{phase: } \arctan(\omega\tau)$$

$$\begin{cases} \omega\tau \ll 1: & j\omega\tau + 1 \cong 1, & \angle 1 = 0^\circ \\ \omega\tau \gg 1: & j\omega\tau + 1 \cong j\omega\tau, & \angle j\omega\tau = 90^\circ \\ \omega\tau \cong 1: & & \angle(j\omega\tau + 1) \cong 45^\circ \end{cases}$$



Summary:

- Class 1: $K_0(j\omega_0)^n$
Class 2: $(j\omega\tau + 1)^{\pm 1}$
- Bode gain-phase relationship
gain: $n \times 20 \text{ dB/decade} \rightarrow$ magnitude through $K_0 \log K_0 |(j\omega)^n| = \log K_0 + n \log(j\omega)$
phase: $n \times 90^\circ$
- First order systems at break point frequency τ
 $\rightarrow$ Asymptotic behaviour used to draw frequency behaviour
- Up next: Class 3 terms & Frequency response of poles and zeros in RHP

evaluate at
 $\omega = 1 \text{ rad/s}$